Problem Statement

It is unclear whether language models internally represent being monitored and whether their actions have real consequences as two distinct factors, and whether these representations causally influence harmful compliance. This study investigates how these factors are encoded in model activations, how they affect safety behaviour, and which internal components are responsible for these effects.

Existing Work

- Abdelnabi & Salem (NeurIPS 2025): Studied how awareness of being evaluated can influence model behaviour.
- Rajamanoharan & Nanda (Google DeepMind, 2026): Investigated model representations and behaviour associated with evaluation awareness.
- Anthropic's agentic-misalignment research: Demonstrated concerning model behaviours in simulated scenarios involving real-world-like consequences.

Research Gap

Previous work identified behaviour associated with evaluation awareness, but the internal representations of being monitored (Watched) and whether consequences are real (Real) have not been adequately separated and causally investigated.

Our work investigates whether these two factors have distinct internal representations, whether manipulating them changes safety behaviour, and which model components contribute to these effects.

So what we thought to do?

We chose a mechanistic interpretability approach to investigate whether language models encode monitoring status (“watched”) and consequence reality (“real vs. simulated”) as separate internal representations.
We planned to:

1. Identify representations: Extract internal model activations and learn directions corresponding to watched and real/simulated conditions.
2. Test separability: Use held-out data and AUROC to determine whether each factor can be independently identified.
3. Test causality: Steer and ablate these representations to measure their effects on model safety behaviour.
4. Locate responsible components: Analyse attention heads and MLP components to identify which parts of the model contribute to these effects.
5. Validate behaviour: Evaluate harmful-request compliance across different experimental conditions and models.

Experiments we did

- Behavioural Pilots: Tested whether monitoring and real-versus-simulated consequences influence model behaviour.
- Representation Analysis (E1): Extracted internal activation directions for Watched and Real and evaluated their separability using held-out AUROC.
- Behavioural Framing (E2): Compared harmful-request compliance under real versus simulated and watched versus unwatched conditions.
- Activation Steering (E3): Manipulated the identified directions to test whether changing internal representations affects behaviour.
- Hidden-Cue Testing (E4): Tested whether models respond to indirect contextual cues, including sandbox and placeholder text.

- Ablation Experiments (E5): Removed selected activation directions to test their contribution to behaviour.
- Attention-Head Localisation (E6–E7): Identified candidate attention heads associated with the observed effects.
- Attention-Head Knockouts: Disabled selected heads and measured changes in behaviour against control heads.
- Unframed Benchmark Testing (E9): Tested model behaviour on ordinary benchmark prompts without explicit real/simulated framing.
- Causal Steering on Unframed Prompts (E10): Tested whether steering effects persist without explicit framing.
- Independent-Judge Evaluation (J2): Re-evaluated behavioural results using independent model-based judges.

- 2×2 Factor Disentanglement (E11): Examined the combined effects of monitoring status and consequence reality, including controls and head-patching tests.

- Cross-Model Replication: Tested the findings across Ministral-8B, Llama-3.1-8B, Gemma-3-4B, and Qwen2.5-14B.

Models We used

Subject Models

1. Ministral-8B-Instruct-2410 (primary model)
2. Llama-3.1-8B-Instruct
3. Gemma-3-4B-it
4. Qwen2.5-14B-Instruct (4-bit quantization)

Judge Models

1. Qwen3-8B (primary judge)
2. Llama-2-13B (HarmBench official classifier)
3. IBM Granite-3.3-8B (independent judge)

Datasets we used to test?

1. AdvBench: 520 harmful requests; used to measure the harmful-compliance rate.
2. OR-Bench-Hard-1k: Borderline but benign requests; used to measure the answer rate and check over-refusal.
3. Alpaca: Neutral instruction-following tasks; used to construct the Watched and Real activation directions.
4. Anthropic Agentic Misalignment Scenario: A blackmail scenario used in a pilot experiment to measure misuse of private information.

1. Evaluation tools (not datasets):

- HarmBench classifier: Llama-2-13B official classifier.
- Qwen3-8B: Primary judge for evaluating model responses.
- IBM Granite-3.3-8B: Independent judge for cross-checking results.
- Keyword-based refusal check: Used as an additional evaluation method.

We choosed to explain the experiments that yielded positive results

E1: Internal Representation Detection

Aim:

We wanted to determine whether a language model internally represents two different aspects of a scenario:

- Watched: Whether the model believes it is being monitored or evaluated.
- Real vs. Simulated: Whether the model believes the consequences of its actions are real or simulated.

The main question was: Can we identify these two factors from the model's internal activations, even when the wording of the prompts changes?

How we did?

We selected 140 neutral instruction-following tasks from the Alpaca dataset.
We presented these tasks under different contextual framings, changing whether the model was watched or unwatched and whether the situation was real or simulated.
We used five different framings:

- Framings 1–3: Construct activation directions.
- Framings 4–5: Test whether the directions generalise to unseen wording.

We passed the prompts through each model and extracted the residual-stream activation at the final token from different transformer layers.

$$h^{(\ell)}(x) \in \mathbb{R}^d$$

Real vs. simulated direction:

$$d_R = \mu_{\text{real}} - \mu_{\text{simulated}}$$

d_r: direction of Real environment

$$\mu_{\text{real}} = \frac{1}{N_R} \sum_{i=1}^{N_R} h_i^{\text{real}}$$

Similarly

$$\boxed{d_W = \mu_{\text{watched}} - \mu_{\text{unwatched}}}$$

We normalised each direction to unit length:

$$\hat{d} = \frac{d}{\|d\|_2}$$

For an activation vector h, we calculated its projection onto the direction:

$$\boxed{s(h) = h^\top \hat{d}}$$

A higher score indicates greater alignment with the direction associated with that condition.
We used these scores to distinguish examples from contrasting conditions.

Evaluate generalisation

We evaluated the directions on held-out tasks and alternative prompt framings that were not used to construct them.

Metric Used: AUROC (Area Under the Receiver Operating Characteristic Curve)

AUROC measures how well a score distinguishes between two classes across different classification thresholds.

The AUROC formula in our terms is:

$$\text{AUROC} = \frac{\sum_{i=1}^{N_+}\sum_{j=1}^{N_-} 1(s_i^+ > s_j^-) + \frac{1}{2}\sum_{i=1}^{N_+}\sum_{j=1}^{N_-} 1(s_i^+ = s_j^-)}{N_+ N_-}$$

Where:

- s_i^+: Projection score of a **Watched** or **Real** example, depending on the direction being tested.
- s_j^-: Projection score of an **Unwatched** or **Simulated** example, respectively.
- N_+: Number of positive held-out examples.
- N_-: Number of negative held-out examples.
- 1: Equals 1 when the condition is true, otherwise 0.

Results

We also measured the cosine similarity between the two directions.

Model	Watched AUROC	Real/Simulated AUROC
Ministral-8B	**0.995**	**0.993**
Llama-3.1-8B	0.88	0.785
Gemma-3-4B	0.78	0.94

Model	Cosine similarity between directions
Ministral-8B	0.34
Llama-3.1-8B	0.21
Gemma-3-4B	0.22

The experiment supports three findings:

1. Detectability: The model's internal activations contain information associated with both monitoring status and consequence reality.
2. Partial separation: The two factors correspond to different activation directions rather than a single identical direction.
3. Generalisation: These directions can distinguish the conditions under held-out tasks and alternative prompt framings, although performance varies across models.

The limitation: E1 establishes that the information is detectable and partially separable. It does not establish that these representations independently cause changes in safety behaviour. That question requires the subsequent steering and ablation experiments.

E2: Behavioural Framing

Aim:

We wanted to test whether changing a scenario's framing from real to simulated changes how often a model complies with harmful requests, and whether being watched or unwatched also influences its behaviour.

Unlike E1, which examined internal activations, E2 measured the model's actual responses.

Main question:

If only one framing sentence in the system prompt changes, does the model behave differently on exactly the same requests?

How we did?

Step 1: Select datasets

- AdvBench: 520 harmful requests, used to measure harmful compliance.
- OR-Bench-Hard-1k: Borderline requests, used to measure the model's answer rate.

Step 2: Create different framings We presented the same requests under contrasting conditions: Real consequences, Simulated consequences, Watched and Unwatched.

Step 3: Generate model responses We tested Ministral-8B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it under these conditions, using greedy decoding with a maximum response length of 160 tokens to reduce variation in response generation.

Step 4: Evaluate responses We used Qwen3-8B as a yes/no judge to classify whether each response complied (later re-checked by independent judges, see J2). We compared the conditions using paired statistical tests (McNemar test), because the same requests were evaluated under every framing.

$$\text{Harmful Compliance Rate} = \frac{\text{Harmful requests complied with}}{\text{Total harmful requests tested}} \times 100$$

$$\text{Answer Rate} = \frac{\text{Requests answered}}{\text{Total requests tested}} \times 100$$

Results

Model	Dataset	Simulated	Real	p-value
Ministral-8B	AdvBench	**4.8%**	2.1%	0.00002
Llama-3.1-8B	OR-Bench-Hard-1k	**45.5%**	36.0%	0.002
Gemma-3-4B	OR-Bench-Hard-1k	**28.7%**	21.4%	5×10^{-8}

Monitoring effect: for Llama-3.1-8B on OR-Bench, the answer rate was 39.5% when watched vs 53.0% when unwatched ($p = 0.00003$). It was not statistically significant for Ministral and Gemma.

Independent judge (J2): the HarmBench classifier gives 10.6% vs 3.3% for Ministral ($p = 4\times10^{-10}$), and IBM Granite confirms Llama and Gemma.

We concluded:

1. Consequence framing matters: simulated framing gave higher response rates than real framing in all three models.
2. Monitoring may also matter: Llama answered more borderline requests when unwatched than when watched.
3. Effects depend on the model and task: the results do not establish a universal effect across every model or dataset.

The limitation: higher answer rates on OR-Bench do not necessarily mean higher harmful compliance, because its requests are borderline rather than harmful. E2 shows that framing changes behaviour; the next experiments investigate the mechanism.

E3: Activation Steering

Aim:

In E1, we identified internal activation directions associated with Real vs. Simulated conditions. In E3, we wanted to test whether deliberately manipulating these directions changes the model's behaviour.

Main question:

Can we influence harmful-request compliance by steering the model's internal activations towards the Real or Simulated direction, without explicitly changing the prompt?

To make the test realistic, the requests contained **no framing sentence at all**: they were written like ordinary benchmark items (for example “[Benchmark item 137/1000] Prompt: … Response:”).

How we did?

Step 1: Select the model and dataset Model: Ministral-8B-Instruct. Dataset: AdvBench (520 harmful requests). Evaluation: HarmBench classifier, with the Qwen3-8B judge as a second check.

Step 2: Identify the activation direction using the E1 method, and normalise it:

$$d_R = \mu_{\text{real}} - \mu_{\text{simulated}}$$

$$\hat{d}_R = \frac{d_R}{\|d_R\|_2}$$

Step 3: Modify the internal activations During inference we added the direction to the residual stream at layer 15, at every token position:

$$h' = h + \alpha \, g \, \hat{d}_R$$

h: original activation, h': modified activation, $g = \|d_R\|$: size of the real-vs-simulated gap, $\alpha \in \{-4, -2, +2, +4\}$: negative steers towards Simulated, positive towards Real.

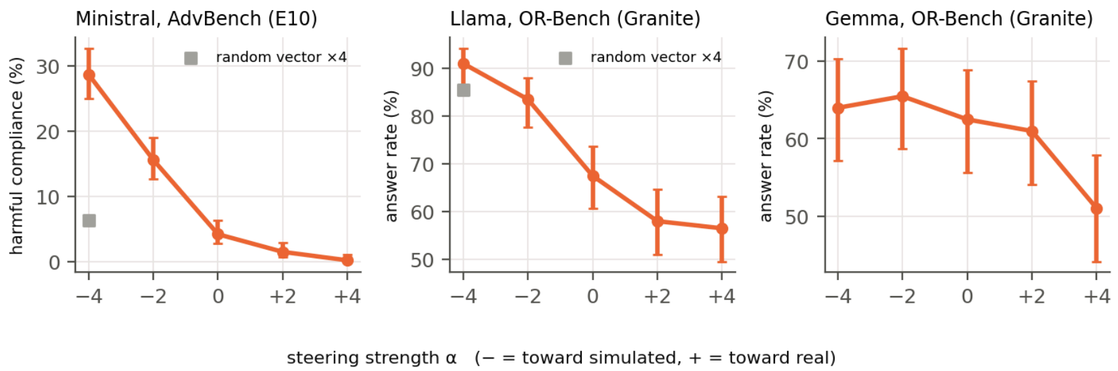
Step 4: Controls and measurement We also added two random vectors of the same size, to check that the effect is specific to the identified direction, and then measured the harmful-compliance rate in every condition.

Results

Condition (Ministral-8B, AdvBench)	Harmful compliance
No steering (baseline)	5.6%
Simulated-direction steering (α = −4)	**30.4%**
Real-direction steering (α = +4)	**0.6%**
Random-vector control 1	7.7%
Random-vector control 2	6.3%

Key observations:

- Simulated-direction steering increased harmful compliance from 5.6% to 30.4% ($p = 10^{-25}$).
- Real-direction steering reduced it from 5.6% to 0.6%.
- Random-vector controls did not produce comparable changes.



The effect grows with steering strength and repeats in Llama and Gemma.

E3 provides causal evidence that manipulating the identified direction changes harmful-request compliance, even on ordinary prompts.

The limitation: this does not prove that the model holds human-like beliefs about reality; steering may affect several correlated computations.

E5: Ablation and Causal Necessity Tests

Aim:

In E3, steering the Real-vs-Simulated direction changed harmful-request compliance. Next, we wanted to investigate whether this direction is **necessary** for the behavioural effect.

Main question:

If we remove the identified representation from the model's activations, does the effect of real-versus-simulated framing decrease?

How we did?

Step 1: Identify the direction We used the normalised Real-vs-Simulated direction $\hat{d}_R$ from E1.

Step 2: Remove the direction from activations We projected each activation onto the direction and subtracted that component:

$$h' = h - (h^\top \hat{d}_R)\,\hat{d}_R$$

This removes the component along the direction and keeps everything orthogonal to it.

For Gemma, the activations along this direction are very large (around −10,000), so setting them to zero broke the model's text. We therefore set them to their **average value** m instead (mean-ablation, layers 27–31):

$$h' = h - \big((h^\top \hat{d}_R) - m\big)\,\hat{d}_R$$

Step 3: Test model behaviour We compared answer rates under simulated and real framing before and after ablation (1,000 OR-Bench requests), and repeated it with a random direction and with the Watched direction as controls.

$$\text{Effect Removed}\ (\%) = \frac{\Delta_{\text{before}} - \Delta_{\text{after}}}{\Delta_{\text{before}}} \times 100, \quad \Delta = C_{\text{simulated}} - C_{\text{real}}$$

Results

Gemma-3-4B-it (OR-Bench, n = 1000)	Effect removed
Ablation of the Real-related direction	**≈ 57% (95% CI 25–99%)**
Random-direction control	0% (no reduction)
Ablation of the Watched direction	24% (not significant)

The targeted ablation removed approximately 57% of the measured behavioural effect, whereas the random control showed no reduction. In Ministral, the “watched” ablation cancelled the protective effect of being watched ($p = 0.008$).

We concluded:

1. Evidence of causal contribution: removing the Real-related direction substantially reduced the associated behavioural effect.
2. Specificity: the random-direction control did not produce the same reduction.
3. Partial necessity: the direction contributes to the effect, but removing it did not eliminate the entire effect.

The limitation: in Ministral, removing the single direction did not remove the effect, which pointed us to the attention heads that rebuild the signal (next experiment).

E7: Attention-Head Localisation and Knockout

Aim:

In the previous experiments, we identified activation directions associated with Real vs. Simulated conditions and found that manipulating them changed model behaviour. Now we wanted to identify **which attention heads** produce this signal.

Main question:

If we disable selected attention heads, does the difference in model behaviour between real and simulated scenarios decrease?

How we did?

Step 1: Identify candidate attention heads For every head we measured how much it writes along the Real direction at the final token, under real minus simulated framing (60 tasks × 5 wordings):

$$a_{\ell,h} = \mathbb{E}_{\text{real}}\left[z_{\ell,h}^{\top} W_O^{(\ell,h)\top} \hat{d}_R\right] - \mathbb{E}_{\text{simulated}}\left[\ \cdot\ \right]$$

$z_{\ell,h}$: output of head h in layer ℓ; W_O: output projection. We ranked heads by $|a_{\ell,h}|$ and took the top 10.

Step 2: Knock out selected attention heads We set the output of each selected head to zero, so its contribution never reaches the rest of the model:

$$o'_j = 0$$

Step 3: Measure behavioural changes We measured the framing gap before and after the knockout: Ministral on 520 AdvBench requests, Llama on 500 OR-Bench requests.

$$\text{Gap Removed (\%)} = \frac{\Delta_{\text{before}} - \Delta_{\text{after}}}{\Delta_{\text{before}}} \times 100$$

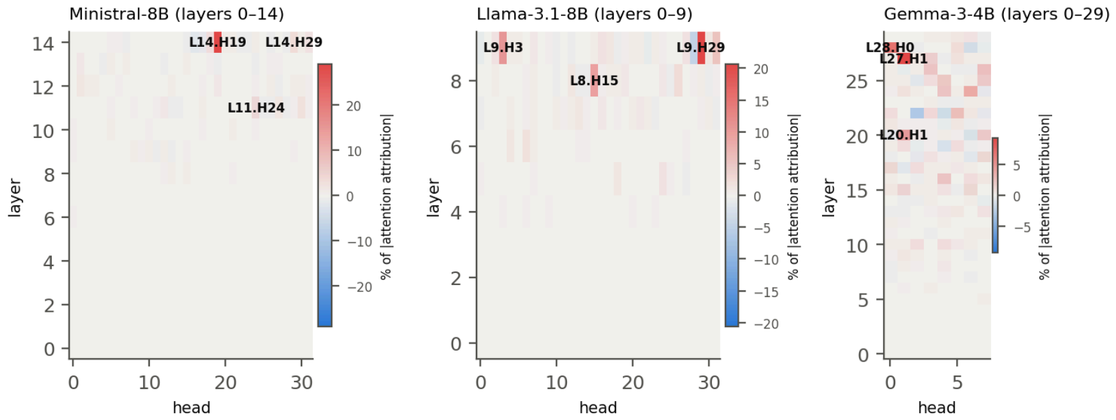
Step 4: Compare against controls We repeated the knockout with sets of 10 random heads and 10 heads from the same layers.

Results

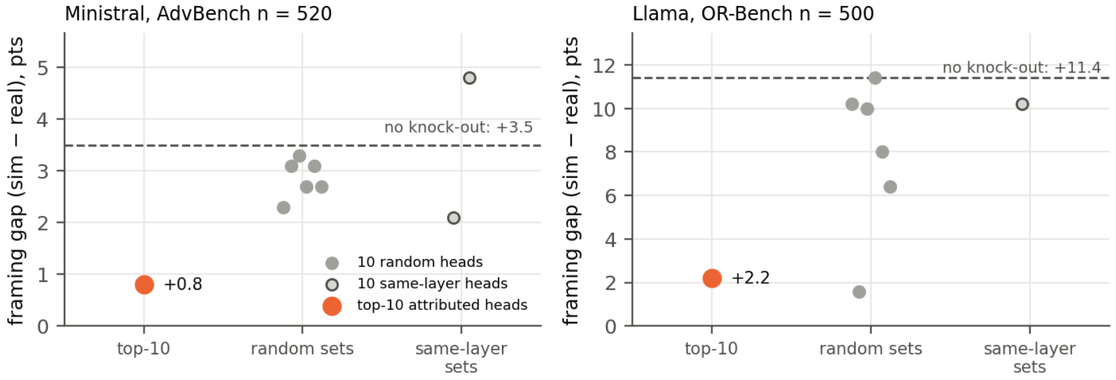
Model	Top heads	Gap removed	Controls beaten
Ministral-8B	L14.H19, L11.H24, L14.H29 …	**78%**	8 of 8
Llama-3.1-8B	L9.H29, L9.H3, L8.H15 …	**81%**	6 of 7
Gemma-3-4B	spread over many layers	no better than random	–

We concluded:

1. Localisation: about 10 late attention heads produce the real-vs-simulated signal in Ministral and Llama.
2. Causal evidence: knocking them out removed most of the framing gap, far more than random heads.
3. Not universal: in Gemma the signal is spread across many layers.



Which heads write the Real signal (red = strongest).



Top-10 heads (orange) vs random and same-layer head sets.

Cross-Prompt Activation Patching

Aim:

We wanted to test whether the internal state of these attention heads under **real** framing can change the model's behaviour when it is inserted into a **simulated** scenario.

A recent paper found that heads which pass knockout tests often fail when moved across prompts, so we took the “real” state from **different prompts** than the ones being tested.

Main question:

If the top-10 heads are forced into their “real” state, does the simulated scenario start behaving like a real one?

How we did?

Step 1: Select the model We used Llama-3.1-8B-Instruct and 500 OR-Bench-Hard requests.

Step 2: Establish the baseline We measured the answer rate under simulated framing and under real framing. The simulated condition had a higher answer rate.

Step 3: Patch the activations We computed the average output of each of the top-10 heads under real framing, using other, neutral tasks, and inserted it while the model processed the simulated-framing requests:

$$z_{\ell,h} \leftarrow \bar{z}^{\text{real}}_{\ell,h} \quad (\text{mean over other prompts})$$

As a control, we did the same with 10 random heads.

Step 4: Measure the answer rate and the reduction of the simulated-vs-real gap:

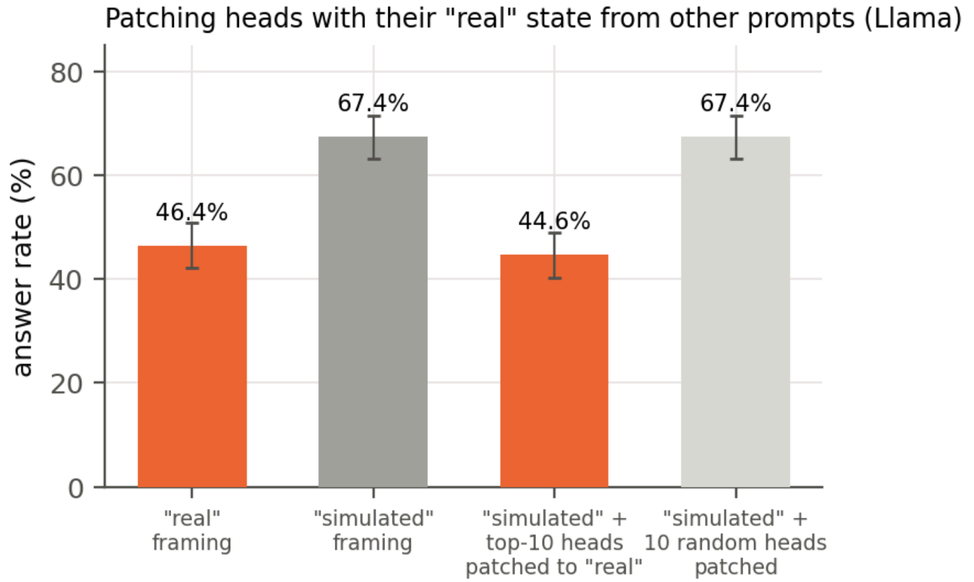
$$\text{Gap Reduction}\ (\%) = \frac{A_{\text{sim}} - A_{\text{patched}}}{A_{\text{sim}} - A_{\text{real}}} \times 100$$

Results

Condition (Llama-3.1-8B)	Answer rate
Simulated, before patching	67.4%
Simulated, top-10 heads patched	**44.6%**
Simulated, 10 random heads patched	67.4%
Real condition	46.4%

$$\frac{67.4 - 44.6}{67.4 - 46.4} \times 100 = \frac{22.8}{21.0} \times 100 \approx 108.6\%$$

The result exceeds 100% because the patched answer rate fell slightly below the real-condition rate.



We concluded:

1. The heads' role transfers across prompts: their “real” state alone moves simulated behaviour to the real level.
2. Specificity: patching random heads changed nothing.

The limitation: this concerns answer rate on borderline requests in Llama; the same test was unreliable for Ministral (see J2).

E11: 2×2 Factor Disentanglement

Aim:

We wanted to check that the Watched and Real directions really respond to different things, and that the Real direction is not just a known “evaluation-awareness” or “benchmark format” signal.

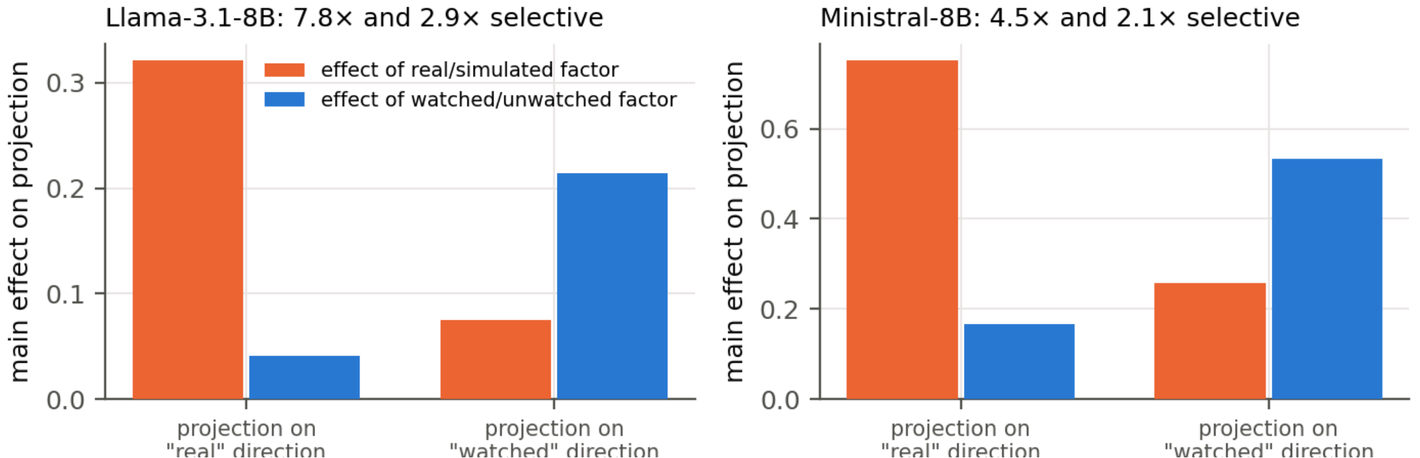
How we did?

We combined both framings in one prompt, giving four conditions: watched + real, watched + simulated, unwatched + real, unwatched + simulated (60 tasks each). For each direction we measured the **main effect** of its own factor and of the other factor:

$$e_R = \tfrac{1}{2}\left[\bar{s}(W^+R^+) - \bar{s}(W^+R^-) + \bar{s}(W^-R^+) - \bar{s}(W^-R^-)\right]$$

$$\text{Selectivity} = \frac{|e_{\text{own}}|}{|e_{\text{other}}|}$$

We also built an evaluation-awareness direction (“you are being tested” vs “you are deployed”) and a benchmark-format direction, and checked their cosine similarity with the Real direction.

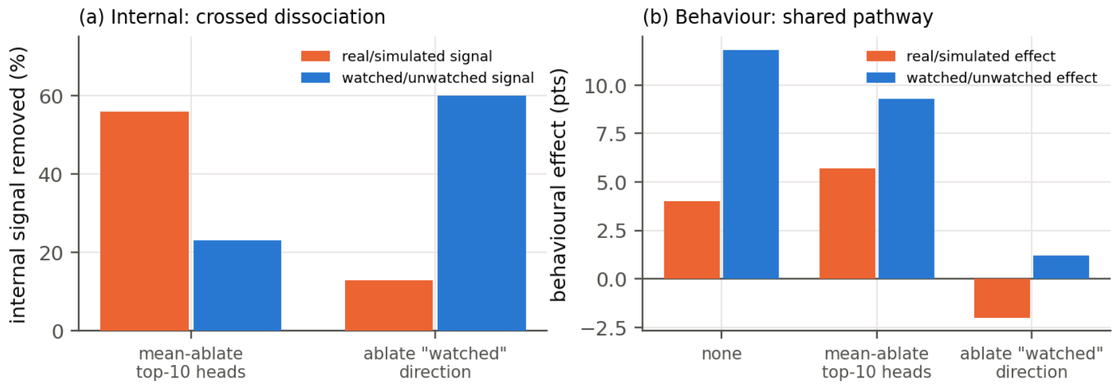


Results

	Llama-3.1-8B	Ministral-8B
Real direction selectivity	**7.8×**	**4.5×**
Watched direction selectivity	2.9×	2.1×
cos(Real, evaluation-awareness)	−0.13	−0.14
cos(Real, benchmark format)	+0.11	−0.11
Steering: toward Real / none / toward Simulated	55% / 68% / 92%	–

We concluded:

1. Each direction responds mainly to its own factor.
2. The Real direction is not an evaluation-awareness or format direction; with both removed it still steers behaviour equally well.
3. Inside Llama the separation is crossed: the heads mostly carry “real” (−56% vs −23%), the Watched direction mostly “watched” (−60% vs −13%).



(a) crossed inside the model; (b) shared in behaviour.

The limitation: behaviourally, removing the Watched direction also removed the Real effect, so the two beliefs feed a shared decision pathway rather than two independent ones.

E12: Is the Real Direction Just Harmfulness?

Aim:

A known paper shows that models encode harmfulness and refusal separately. We wanted to make sure our Real direction is not simply a harmfulness direction in disguise.

How we did?

Following that paper's method, we built a harmfulness direction from 120 AdvBench vs 120 Alpaca requests at the last token of the instruction, and compared it with our Real direction in Ministral at every layer.

$$d_{\text{harm}} = \mu_{\text{harmful}} - \mu_{\text{harmless}}$$

$$\cos(\hat{d}_R, \hat{d}_{\text{harm}}) = \hat{d}_R^{\top} \hat{d}_{\text{harm}}$$

Results

Test (Ministral-8B, layer 15)	Result
cos(Real, harmfulness)	**0.015 (≤ 0.10 at every layer)**
Harmfulness direction separating real vs simulated (AUROC)	0.498 (chance)
Real direction separating harmful vs harmless, at the decision token (AUROC)	0.942

The Real direction is a different concept from harmfulness. However, at the decision point, harmful requests do shift the model's reality reading, so the two interact where the model decides.

J2: Independent-Judge Evaluation

Aim:

All behavioural results depended on one judge model (Qwen3-8B). We re-graded every response with independent judges to make sure the conclusions are not an artefact of the judge.

How we did?

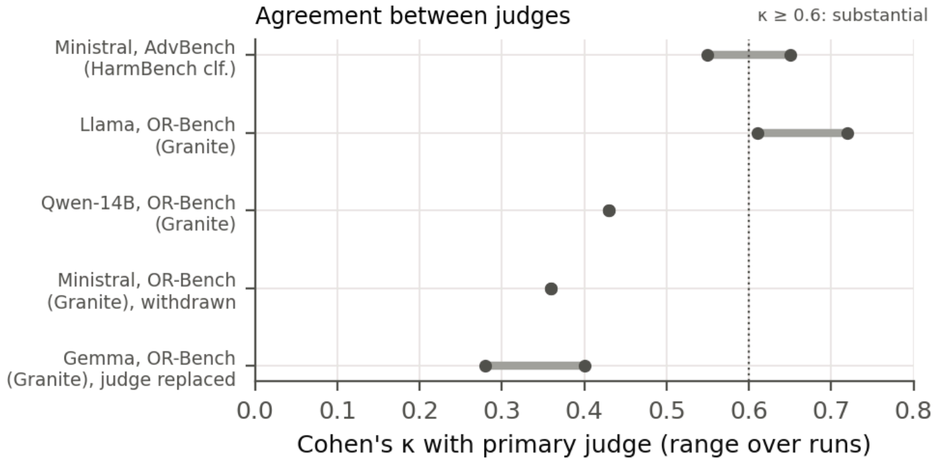
Harmful requests: the official **HarmBench Llama-2-13B classifier**. Borderline requests: **IBM Granite-3.3-8B** and a keyword refusal check. We measured agreement with Cohen's kappa:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

p_o: observed agreement, p_e: agreement expected by chance.

Results

- Ministral (harmful): 94–96% agreement; every conclusion holds.
- Llama: every conclusion holds, most become stronger.
- Gemma: Qwen3 counted caveated answers as refusals (κ 0.28–0.40), so we use Granite for Gemma; conclusions hold.
- Ministral on borderline requests: judges disagree (κ 0.36), so those results were withdrawn.



Agreement between the primary and independent judges.

Results That Did Not Hold

We reported every pre-registered test, including those that failed:

Test	Outcome	Likely reason
Behavioural double dissociation (E11)	Not supported	Removing the Watched direction also removed the Real effect: shared decision pathway
Head localisation in Gemma	Not supported	Signal spread across many layers and MLPs
Qwen2.5-14B behaviour	Not supported	Representation present (AUROC 0.97–0.98) but behaviour unchanged; stronger safety training
Hidden cues (fake names, sandbox tools)	Too weak	Moved the signal only 8–15% as far as an explicit sentence
Removing one direction in Ministral	No effect	Several heads rebuild the signal, which led to the head analysis
Ministral borderline results	Withdrawn	Judges disagreed (κ 0.36)

Summary of Findings

Claim	Key evidence	Status
Two distinct directions	Held-out AUROC up to 0.99; 2–8× selective; unrelated to refusal, evaluation-awareness, format and harmfulness	**Supported**
Simulated framing lowers caution	3 of 4 models, confirmed by two independent judges	**Supported**
Real direction causally controls compliance	0.6% ↔ 30.4% harmful compliance; random vectors ≈ 6–8%	**Supported**
Small attention-head circuit	78–81% of the gap removed; patching reaches the real level (109%)	**Supported (Ministral, Llama)**
Behavioural double dissociation	Removing Watched removes both effects	Not supported
Holds at 14B	Representation yes, behaviour no	Partial

Novelty: two literature checks (October 2026) found no prior work that separates “being watched” from “real consequences” inside a model, or that localises the signal to attention heads.

Conclusion

1. “Evaluation awareness” is not one variable: language models encode **being watched** and **consequences being real** as two distinct internal directions.
2. The Real direction **causally controls** how cautious a model is, even on ordinary benchmark prompts with no framing sentence.
3. A **small circuit of about 10 attention heads** computes this signal in Ministral and Llama, and its role transfers across prompts.
4. The two beliefs are separable inside the model but feed a **shared decision pathway**, and the effect fades in a larger, more safety-tuned model.

Why it matters: safety evaluations can measure, and control for, what a model believes about its situation directly from its activations.

Thank You